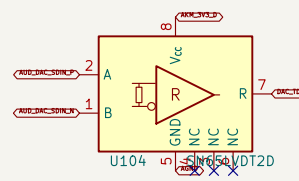
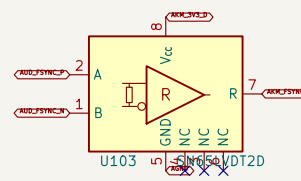
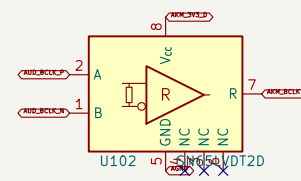
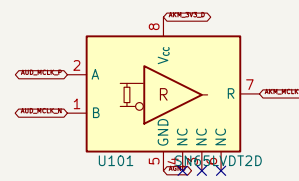
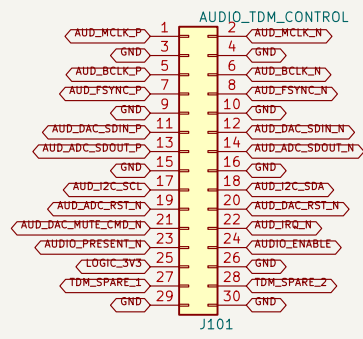


1. LOCKED CARRIER HARNESS

2. TERMINATED LVDS RECEIVERS



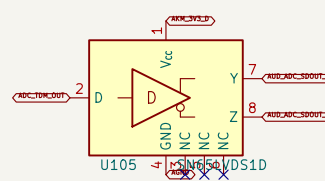
MCLE: integrated I/O elem. terminals

ECLE: Integrated I/O elem. termination

PS100C: Integrated 100 ohm termination

IAC_5009: Integrated I/O pin termination

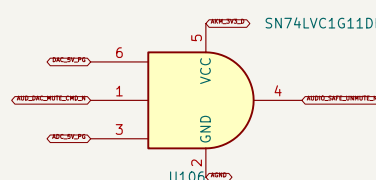
3. ADC RETURN LVDS DRIVER



4. CONTROL DEFAULTS AND POWER STATE



5. HARDWARE FAIL—SILENT GATE



100 ohm differential; interleaved returns; no branch stubs
87832–6423 pin assignment is locked and matches CM5–CARRIER J901
One LVDS driver returns the ADC TDM data stream to the CM5 carrier
Four terminated LVDS receivers carry MCLK, BCLK, FSYNC, and DAC TDM data

ProComm
Sheet: /TDM / Clock / Control/
File: Audio-TDM-Clock.kicad_sch

Title: ProComm AUDIO-8X8 - TDM, Clock, and Control Interface

1. AK5558VN EXACT PIN CAPTURE

2. MODE STRAPS

Accession	Gene	Protein	Accession	Gene	Protein
U00001	oriC	oriC	U00001	oriC	oriC
U00002	oriT	oriT	U00002	oriT	oriT
U00003	oriV	oriV	U00003	oriV	oriV
U00004	oriW	oriW	U00004	oriW	oriW
U00005	oriX	oriX	U00005	oriX	oriX
U00006	oriY	oriY	U00006	oriY	oriY
U00007	oriZ	oriZ	U00007	oriZ	oriZ
U00008	oriA	oriA	U00008	oriA	oriA
U00009	oriB	oriB	U00009	oriB	oriB
U00010	oriC	oriC	U00010	oriC	oriC
U00011	oriD	oriD	U00011	oriD	oriD
U00012	oriE	oriE	U00012	oriE	oriE
U00013	oriF	oriF	U00013	oriF	oriF
U00014	oriG	oriG	U00014	oriG	oriG
U00015	oriH	oriH	U00015	oriH	oriH
U00016	oriI	oriI	U00016	oriI	oriI
U00017	oriJ	oriJ	U00017	oriJ	oriJ
U00018	oriK	oriK	U00018	oriK	oriK
U00019	oriL	oriL	U00019	oriL	oriL
U00020	oriM	oriM	U00020	oriM	oriM
U00021	oriN	oriN	U00021	oriN	oriN
U00022	oriO	oriO	U00022	oriO	oriO
U00023	oriP	oriP	U00023	oriP	oriP
U00024	oriQ	oriQ	U00024	oriQ	oriQ
U00025	oriR	oriR	U00025	oriR	oriR
U00026	oriS	oriS	U00026	oriS	oriS
U00027	oriT	oriT	U00027	oriT	oriT
U00028	oriU	oriU	U00028	oriU	oriU
U00029	oriV	oriV	U00029	oriV	oriV
U00030	oriW	oriW	U00030	oriW	oriW
U00031	oriX	oriX	U00031	oriX	oriX
U00032	oriY	oriY	U00032	oriY	oriY
U00033	oriZ	oriZ	U00033	oriZ	oriZ
U00034	oriA	oriA	U00034	oriA	oriA
U00035	oriB	oriB	U00035	oriB	oriB
U00036	oriC	oriC	U00036	oriC	oriC
U00037	oriD	oriD	U00037	oriD	oriD
U00038	oriE	oriE	U00038	oriE	oriE
U00039	oriF	oriF	U00039	oriF	oriF
U00040	oriG	oriG	U00040	oriG	oriG
U00041	oriH	oriH	U00041	oriH	oriH
U00042	oriI	oriI	U00042	oriI	oriI
U00043	oriJ	oriJ	U00043	oriJ	oriJ
U00044	oriK	oriK	U00044	oriK	oriK
U00045	oriL	oriL	U00045	oriL	oriL
U00046	oriM	oriM	U00046	oriM	oriM
U00047	oriN	oriN	U00047	oriN	oriN
U00048	oriO	oriO	U00048	oriO	oriO
U00049	oriP	oriP	U00049	oriP	oriP
U00050	oriQ	oriQ	U00050	oriQ	oriQ
U00051	oriR	oriR	U00051	oriR	oriR
U00052	oriS	oriS	U00052	oriS	oriS
U00053	oriT	oriT	U00053	oriT	oriT
U00054	oriU	oriU	U00054	oriU	oriU
U00055	oriV	oriV	U00055	oriV	oriV
U00056	oriW	oriW	U00056	oriW	oriW
U00057	oriX	oriX	U00057	oriX	oriX
U00058	oriY	oriY	U00058	oriY	oriY
U00059	oriZ	oriZ	U00059	oriZ	oriZ
U00060	oriA	oriA	U00060	oriA	oriA
U00061	oriB	oriB	U00061	oriB	oriB
U00062	oriC	oriC	U00062	oriC	oriC
U00063	oriD	oriD	U00063	oriD	oriD
U00064	oriE	oriE	U00064	oriE	oriE

AK5558V
U201

3. REFERENCES AND LOCAL BYPASS

Steps establish the safe hardware default: firmware still writes and verifies the serial-control registers after reset

Y91F pair:

VOX/ pair

YSD pair:

VGLF pair 4

4.7uF
C281

100nF
C284

10uF
C285

QFN exposed-pad geometry and assembly coupon remain a routing release gate
Each VREFH uses 20 ohm plus 100 nF and 100 uF to its VREFL/AGND
AVDD=5.0 V, TVDD=3.3 V, internal 1.8 V LDO enabled
TDM256, 32-bit I2S, slave mode: TDM1:0=10, DIF1:0=11, MSN=0

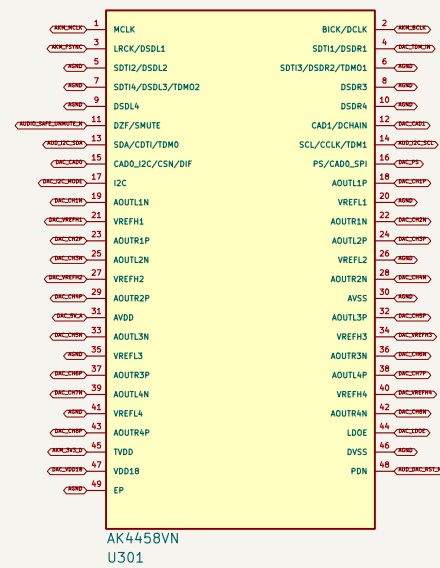
ProComm
Sheet: /AK5558VN ADC/

Title: ProComm AUDIO-8X8 – AK5558VN Eight-Channel ADC

Size: A1	Date: 2026-08-16	Rev: A1
KiCad: EDA-10.0.5-rc1		Id: 3/2

1. AK4458VN EXACT PIN CAPTURE

2. SERIAL-CONTROL DEFAULTS



CR01:B+01 selects DAC address 0x11; AOC remains at 0x10. TIM256 and 32-bit IQ5 are written only while route is asserted.

The unused S0TQ/3y4 and OSD pins are tied low in PCM TDM mode to prevent floating inputs.

3. REFERENCES AND LOCAL BYPASS

Every ILR pin 4 and shell leads to XLR/CRISIS at the connector. GND and shield meet only at the controlled board-entry point.



EIGHT IDENTICAL OUTPUT CHANNELS – SIGNAL FLOWS LEFT TO RIGHT

ANALOG → 120 MHz/200MHz/400MHz/800MHz → 40 pins → 100MHz → 100 MHz clock input → 100MHz/200MHz/400MHz/800MHz → 40 pins

Keep two channels only when at least 10 channels, 100 channels, 200 channels, 400 channels, 800 channels, 1600 channels, 3200 channels, 6400 channels, 12800 channels, 25600 channels, 51200 channels, 102400 channels, 204800 channels, 409600 channels, 819200 channels, 1638400 channels, 3276800 channels, 6553600 channels, 13107200 channels, 26214400 channels, 52428800 channels, 104857600 channels, 209715200 channels, 419430400 channels, 838860800 channels, 1677721600 channels, 3355443200 channels, 6710886400 channels, 13421772800 channels, 26843545600 channels, 53687091200 channels, 107374182400 channels, 214748364800 channels, 429496729600 channels, 858993459200 channels, 1717986918400 channels, 3435973836800 channels, 6871947673600 channels, 13743895347200 channels, 27487790694400 channels, 54975581388800 channels, 109951162777600 channels, 219902325555200 channels, 439804651110400 channels, 879609302220800 channels, 1759218604441600 channels, 3518437208883200 channels, 7036874417766400 channels, 14073748835532800 channels, 28147497671065600 channels, 56294995342131200 channels, 112589990684262400 channels, 225179981368524800 channels, 450359962737049600 channels, 900719925474099200 channels, 1801439850948198400 channels, 3602879701896396800 channels, 7205759403792793600 channels, 14411518807585587200 channels, 28823037615171174400 channels, 57646075230342348800 channels, 115292150460684697600 channels, 230584300921369395200 channels, 461168601842738790400 channels, 922337203685477580800 channels, 1844674407370955161600 channels, 3689348814741910323200 channels, 7378697629483820646400 channels, 14757395258967641292800 channels, 29514790517935282585600 channels, 59029581035870565171200 channels, 118059162071741130342400 channels, 236118324143482260684800 channels, 472236648286964521369600 channels, 944473296573929042739200 channels, 1888946593147858085478400 channels, 3777893186295716170956800 channels, 7555786372591432341913600 channels, 15111572745182864683827200 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278759314981632789269196478408104518824755200 channels, 557518629963265578538392956816209037649510400 channels, 1115037259926531157076785913632418075299020800 channels, 2230074519853062314153571827264836150598041600 channels, 446014903970612462

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1. AUDIO_12V ENTRY AND SINGLE GROUND STAR

3. QUIET 5.5 V PRE-RAIL

2. ISOLATED BIPOLAR LINE-STAGE RAILS

4. SEPARATE ADC AND DAC 5 V LT3045 RAILS

5. AKM DIGITAL RAIL AND 2.5 V INPUT COMMON MODE

6. ERC SOURCES AND SEQUENCING CONTRACT